

Mohammad Taher Ahmadi

Game Engine Developer | Graphics Programmer



Summary

Game developer with 11+ years of experience specializing in game engine architecture, graphics programming, and performance-critical systems. Experienced in building custom engines (C++, C#, ECS-based), implementing physics systems from scratch, and working with low-level rendering techniques (OpenGL, shaders, ray marching). Strong background in both shipping live games and developing scalable engine tools, with a focus on clean architecture and data-oriented design.



Education

- Bachelor's Degree (In Progress): Computer Engineering | Iran University of Science and Technology (IUST)



Featured Engine & Graphics Projects

Enigma 2D Game Engine (C#, OpenTK, ECS)

- Built a custom 2D game engine with integrated WPF editor
- Implemented Entity Component System (ECS) architecture
- Developed physics engine from scratch (collision detection, rigidbody dynamics)
- Designed modular, scalable architecture
- Created full documentation website
- Links: [Github](#) | [Documentation](#)

C++ Game Engine (SDL, ImGui, Lua)

- Developed a C++ game engine with Lua scripting integration
- Implemented rendering and tooling using SDL and ImGui
- Focused on extensibility and scripting-driven workflows
- Link: [Github](#)

The Blue Ice (Ray Marching / Fractal Game)

- Developed a PC game using ray marching and fractal rendering
- Focused on procedural visuals and non-rasterized rendering techniques
- Applied advanced shader-based rendering approaches
- Link: [Game](#)



Technical Highlights

- Built ECS-based game engine from scratch
- Implemented custom physics engine
- Developed ray-marching rendering project
- Experience with live multiplayer systems and backend integration



Professional Experience

Unity Developer / Meta Systems Engineer

Medrick – Mafia Nights / Zodiac / Secret Hitler

July 2023 – September 2024

- Developed multiplayer social-deduction games for live market
- Designed and implemented meta systems (profiles, progression, social features)
- Integrated backend services using REST APIs and WebSockets
- Worked on real-time systems (chat, matchmaking, group features)
- Maintained and optimized live production systems
- Technologies: Unity, C#, FastAPI, WebSockets

Contacts

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Tech Skills & Tools

- Engine & Systems**
 - Architecture: Entity Component System (ECS), Data-Oriented Design (DoD), modular engine design
 - Core Systems: scene management, asset/resource pipelines, serialization, scripting integration (Lua)
 - Concurrency: Unity Job System, basic multithreading patterns
 - Tooling: custom editors (WPF), engine tools, debugging & profiling utilities
- Graphics & Rendering**
 - APIs/Libraries: OpenGL (via OpenTK), SDL
 - Rendering Pipeline: batching, draw call optimization, frustum & occlusion culling, LOD systems
 - Shaders: HLSL, GLSL, Compute Shaders (optimization & debugging)
 - Techniques: ray marching, fractal rendering, basic ray tracing, post-processing
- Physics & Mathematics**
 - Collision Detection: AABB, OBB, SAT (broadphase & narrowphase)
 - Dynamics: rigidbody simulation, force integration
 - Math: linear algebra (vectors, matrices), transforms, projections
- Programming**
 - Languages: C++ (engine development), C# (Unity & tools), Python (tooling)
 - Practices: SOLID, design patterns, memory management, performance optimization

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Game developer

Game Developer

Atieh Dadeh Pardaz – Web Mini-Games Platform

Dec 2024 – Feb 2025

- Developed multiple web-based mini-games using Phaser
- Focused on clean architecture and maintainable systems
- Integrated backend services and real-time communication
- Technologies: JavaScript, Phaser, REST API

Python Instructor

Medrick Academy

Nov 2024 – Mar 2025

- Taught Python fundamentals with focus on clean code and problem solving
- Mentored students in practical programming concepts

Unity Developer (Localization)

Medrick FZE – Merge Decor

- Localized and adapted a published game for regional markets
- Ensured UI and gameplay consistency across languages
- [Link](#)

Selected Projects

DecorChi (Published Mobile Game)

- Designed and developed a match-3 mobile game from concept to release
- Published on CafeBazaar
- [Link](#)

Tiara and Her Troubled Pedigree

- Award-winning serious game focused on cognitive improvement
- Received Fajr Festival Diploma
- Referenced in academic research
- [Link](#)

Simple Renderer & Ray Tracing Projects

- Built a simple 3D renderer
- Implemented ray tracing project in C++
- Link: [Github](#)

Additional Projects

Collection of indie and game jam projects focused on:

- Rendering and shader optimization
- AI systems and gameplay architecture
- Performance optimization and memory management
- Physics-based gameplay

Available at: tch.io

What I'm Looking For

Looking for a role in a team oriented environment where I can learn and grow in game development every day, with exposure to graphics programming, algorithms, gameplay systems, and core mechanics. I'm motivated by deep technical learning and hands-on problem solving, and I aim to continuously improve both my skills and the quality of the games I work on.

Thank you for taking the time to review my resume

Networking & Backend

- Concepts: client-server architecture, synchronization, real-time systems
- Technologies: REST APIs, WebSockets, FastAPI
- Tools & Ecosystem**
 - Engines/Frameworks: Unity (Addressables, scene streaming), ImGui
 - Version Control: Git (Git Flow), basic CI/CD
 - Other: Blender (basic modeling, rigging, animation)

Soft Skills

- Strong problem-solving mindset
- Fast learner, self-driven
- Collaborative and adaptable

Languages

- Persian
- English
- Japanese